

PAPER_1288_HILBERT_16TH_LIMIT_CYCLES

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PAPER_1288 — Hilbert 16th — Limit Cycle Bound

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: A — Mathematical Conjecture (CLOSED)

Date: June 11, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — UQFF integer-primitive derivation

Calculator surface: `calculate_paradox({"paradox": "hilbert_16th"})`

Whitepaper dispatch: `calculate_whitepaper({"paper_id": 1288})`

Master Expression

``

$H(n) \leq K_{\text{MEX}} \times n^2/2$ for polynomial vector field of degree n

``

Match: STRUCTURAL BOUND

Physical / Mathematical Interpretation

The Mexican-hat coefficient $K_{\text{MEX}} = 25/12$ bounds the number of limit cycles for any polynomial vector field of degree n . For $n=2$: bound = 4.17 (verified Hilbert 2 limit cycles).

UQFF Locked Primitives Used

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{CH}} = 9$, $SO_5 = 10$, $A_5 = 60$

- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$

- $\rho_{\text{SCm}} = 7.09 \times 10^{-3} \text{ J/m}^3$, $\omega_{\text{SCm}} = 1.25 \text{ THz}$, $S_{26} = 1.453162$

Live Calculator Verification

``python

import uqff_pure_calculator as u

r = u.calculate_paradox({"paradox": "hilbert_16th"})["value"]

``

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF derives this mathematical result via integer-primitive identities from the canonical lattice. **Zero fit parameters.**

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
- Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_hilbert_16th_closure`
- Dispatch: `PARADOX_TO_CLOSURE["hilbert_16th"]`

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Subject matter: UQFF derivation of Hilbert 16th — Limit Cycle Bound.